

LASTMILE GIG

AI SINGULARITY OPERATING SYSTEM
FOR SOUTH AFRICA'S LAST-MILE DELIVERY INDUSTRY

MASTER DEVELOPMENT PLAN · v1.0 · 2026

lastmilegig.aagais.co.za

21

MODULES

810K

LINES OF CODE

300+

DEV PHASES

19

TECH DOCS

1. Executive Overview

Lastmile Gig is the AI Singularity Operating System for South Africa's last-mile delivery industry.

Lastmile Gig (Pty) Ltd is a technology subsidiary of the Affiliated Architecture Group (AAG), headquartered in KwaZulu-Natal, South Africa. The platform is architected as a full-stack, blockchain-anchored, IoT-integrated, AI-first operating system serving five primary market segments: gig drivers, corporate fleet clients, restaurant and hospitality partners, enterprise logistics operators, and development finance investors.

The platform is built on a polyglot microservices architecture deploying TypeScript/NestJS, Python/FastAPI, Go, Rust, Java/Spring Boot, and Elixir/Phoenix across 21 distinct functional modules — all hosted on AWS with Terraform-provisioned infrastructure, Kubernetes orchestration via AWS EKS, and a global top-1% observability stack comprising Grafana, Prometheus, OpenTelemetry, Loki, Tempo, Sentry, Datadog, and PagerDuty.

METRIC	VALUE	METRIC	VALUE
Platform Modules	21	Estimated Total LoC	~810,000
Development Phases	300+	Technical Documents	19 Markdown Files
Backend Languages	6 (Polyglot)	Cloud Provider	AWS (Fully)
Database Layer	Supabase + Atlas + Upstash	AI/Agent Frameworks	LangChain, LangGraph, CrewAI
Blockchain	Polygon CDK (L2)	ESG Target	30% EV Fleet
Payment Gateways	7 Gateways	Compliance	POPIA + GDPR

2. 21 Platform Modules

Complete module inventory with technology allocation.

#	MODULE	TECH LAYER	STATUS
01	Corporate Landing & Investor Relations	Next.js 14, Sanity CMS, Auth0, Power BI	PLANNED
02	AI Orchestration Platform	Python/FastAPI, LangChain, CrewAI, Kafka (MSK)	PLANNED
03	Driver Rental Module	Angular 17, NestJS, Paystack, TimescaleDB (IoT)	PLANNED
04	Contracted Fleet Module	Angular 17, Go, Polygon CDK, Supabase	PLANNED
05	Branded Fleet Module	Next.js, Java/Spring Boot, Stripe, MongoDB Atlas	PLANNED
06	Fleet Servicing Hub	Rust, TimescaleDB, IoT Sensors, Bosch/SupaQuick APIs	PLANNED
07	Customer Ordering App	React Native/Expo, NestJS, AWS Rekognition	PLANNED
08	Blockchain Layer	Rust, Polygon CDK, Hardhat, ethers.js, The Graph	PLANNED
09	Security Layer	Auth0, HashiCorp Vault, AWS WAF, Snyk, SonarQube	PLANNED
10	Enterprise Logistics	Java/Spring Boot, Go, Kafka/Flink, Takealot/DSV APIs	PLANNED
11	ESG Framework	Angular, Power BI, CrewAI ESG Agents, TimescaleDB	PLANNED
12	APIs & Integrations Hub	AWS API Gateway, NestJS, OpenAPI, ERP/POS adapters	PLANNED
13	AI & Data Intelligence	Python, SageMaker, LangGraph, Pinecone, Elasticsearch	PLANNED
14	Admin & Compliance	Angular, NestJS, Supabase/PostgreSQL, RBAC engine	PLANNED
15	Restaurant Storefronts	Next.js, Sanity, Cloudinary, Elixir/Phoenix	PLANNED
16	Driver Earnings & Wallet	Angular, Java, Paystack, Ozow, PDF tax certs	PLANNED
17	Insurance & Risk Module	NestJS, Go, LangGraph, Naked/Guardrisk APIs	PLANNED
18	Customer Loyalty & Rewards	Next.js, Upstash Redis, LangChain RAG, BullMQ	PLANNED
19	Command Centre	Angular, Go, Elixir Phoenix, Grafana, live map	PLANNED
20	Developer Portal & Public API	NestJS, OpenAPI/Swagger, AWS API Gateway, Sandbox	PLANNED
21	Communications Hub	Twilio, Firebase, SendGrid, Elixir Phoenix Channels	PLANNED

3. Polyglot Technology Stack

Full specification of all languages, frameworks, services, and tools.

3.1 Backend Language Allocation

LANGUAGE/TECH	SERVICE ALLOCATION	RATIONALE
TypeScript + NestJS	API Gateway, Corporate, Partner, Compliance, Comms Services	Type-safe, structured, scales well for microservices
Python + FastAPI	AI/ML inference, LangChain/LangGraph/CrewAI agent orchestration	Dominate AI/ML ecosystem; async-native
Go	Dispatch engine, real-time tracking, Kafka consumers, IoT gateways	Extreme throughput, low latency, concurrency model
Rust	Cryptographic layer, blockchain interface, telemetry ingestion	Memory safety, zero-cost abstractions, performance
Java + Spring Boot	Enterprise logistics, ERP/POS integration, billing engine	Enterprise integration patterns, SAP/Sage compatibility
Elixir + Phoenix	WebSocket real-time services, driver location, live order tracking	Truly M concurrency — millions of persistent connections

3.2 Frontend Architecture

FRAMEWORK	MODULES ALLOCATED	DESIGN SYSTEM
Next.js 14 (App Router)	Corporate Landing (M1), Restaurant Storefronts (M15), Customer Ordering App web, Loyalty Module (M18)	Tailwind CSS + shadcn/ui
Angular 17	Orchestration (M2), Fleet Modules (M3/M4), Admin (M14), ESG (M11), Command Centre (M19), Driver Wallet (M16)	Angular Material + custom DS
React Native + Expo	Customer Mobile App (M7), Driver Mobile App	Custom RN component library

3.3 Database Architecture

DATABASE	TYPE	SERVICE	USE CASE
PostgreSQL	Relational	Supabase (hosted)	Primary data store, auth, realtime subscriptions
Redis	Cache / Queue	Upstash (serverless)	Session cache, rate limiting, BullMQ job queues
MongoDB	Document Store	MongoDB Atlas	Delivery event logs, menu data, unstructured records
TimescaleDB	Time-Series	Supabase extension	IoT telemetry, fleet health, carbon tracking
Elasticsearch	Search Engine	AWS OpenSearch	Menu search, driver search, full-text queries
Pinecone	Vector DB	Pinecone Cloud	RAG pipelines, semantic search, LLM embeddings

4. AI & Agentic Intelligence Layer

Multi-framework agentic architecture using LangChain, LangGraph, and CrewAI.

FRAMEWORK	ROLE	AGENTS / PIPELINES
LangChain	LLM orchestration, RAG pipelines, document processing	Partner onboarding agent, Menu extraction, Driver Q&A
LangGraph	Stateful multi-step agent workflows, decision graphs	Dispatch decision agent, Fraud investigation workflow, Risk scoring
CrewAI	Multi-agent crews for complex coordinated tasks	Demand forecasting crew, Route optimisation crew, ESG reporting crew
AWS Bedrock	Foundation model hosting (Claude, Titan)	Primary LLM backbone for all production agentic services
AWS SageMaker	Custom model training and inference pipelines	Driver performance scoring, demand forecasting, anomaly detection
Pinecone	Vector database for RAG and semantic retrieval	Menu search, driver profile matching, historical delivery RAG

Human-in-the-Loop (HITL) Agentic Development

All agentic development on the Lastmile Gig platform follows a Human-in-the-Loop (HITL) protocol. Agent workflows require human approval gates at critical decision nodes — dispatch overrides, fraud escalations, contract generation, and ESG reporting sign-off. The 19 Markdown documents produced as part of this master plan serve as the canonical ruleset and standards specification for all agentic development teams.

5. Infrastructure, Cloud & DevOps

Fully AWS, Terraform-provisioned, Kubernetes-orchestrated, GitOps-delivered.

SERVICE	ROLE
AWS EKS	Production Kubernetes cluster — container orchestration for all microservices
Terraform + Terragrunt	All infrastructure provisioned as code; DRY multi-environment management
Helm Charts	Kubernetes package management for all service deployments
ArgoCD	GitOps continuous delivery — all deployments declarative via Git
Coolify on AWS EC2	Internal tool hosting, staging environments, developer preview deployments
Istio Service Mesh	mTLS between services, traffic management, circuit breaking, observability
Karpenter	Intelligent node autoscaling on EKS based on real-time workload demand
AWS MSK (Kafka)	Managed Apache Kafka for event streaming backbone across all services
AWS CloudFront + S3	Global CDN for static assets, media, and frontend delivery
AWS Route 53	DNS management — lastmilegig.aagais.co.za and all subdomains
AWS Secrets Manager	Centralised secrets management for all service credentials
AWS CloudTrail	Full audit trail of all API calls and infrastructure changes
GitHub Actions	CI pipelines — build, test, lint, security scan on every pull request
Checkov	Infrastructure-as-Code security scanning on all Terraform plans

6. Observability Stack — Global Top 1%

Full-stack observability covering metrics, logs, traces, alerting, and chaos engineering.

TOOL	CATEGORY	PURPOSE
Grafana	Dashboards	Unified visualisation of all metrics, logs, and traces
Prometheus	Metrics	Time-series metrics collection from all services
OpenTelemetry	Instrumentation	Vendor-neutral traces, metrics, logs across all 6 languages
Loki	Log Aggregation	Centralised log storage and querying (Grafana native)
Tempo	Distributed Tracing	Trace every request across all microservices
Jaeger	Distributed Tracing	Complementary trace analysis and root-cause debugging
Sentry	Error Tracking	Frontend + backend exception capture and alerting
Datadog	APM / Infra	Application performance monitoring and infrastructure monitoring
PagerDuty	Incident Management	On-call scheduling, alert routing, incident response workflows
AWS CloudWatch	Cloud Monitoring	Native AWS resource monitoring and log groups
AWS Fault Injection	Chaos Engineering	Controlled fault injection for resilience testing

7. Payment Ecosystem

Seven payment gateways covering South Africa, Pan-Africa, international, and crypto settlement.

GATEWAY	TYPE	MARKET	USE CASE
Paystack	Card / EFT / Mobile	South Africa + Nigeria	Primary ZA gateway — driver payouts, customer payments
Stripe	Card / SEPA / ACH	International	Corporate client billing, SaaS subscriptions
Flutterwave	Multi-method	Pan-African (30+ countries)	Pan-African expansion — future market entry
Peach Payments	Bank-grade gateway	South Africa	High-value corporate transactions, enterprise billing
Ozow	Instant EFT	South Africa	Driver instant payouts, low-fee bank transfers
SnapScan / MoMo	Mobile payments	South Africa	Customer QR code payments, in-person orders
Polygon CDK	Crypto / Smart Contract	Blockchain	Driver smart contract payouts, partner SLA settlements

8. Lines of Code Estimation

Cumulative LoC estimate across all layers of the Lastmile Gig ecosystem.

LAYER	TECHNOLOGIES	ESTIMATED LoC
Frontend — Next.js surfaces	TypeScript, React, Tailwind, shadcn/ui	~75,000
Frontend — Angular surfaces	TypeScript, Angular 17, Angular Material	~65,000
Mobile Apps (Customer + Driver)	React Native, Expo, TypeScript	~40,000
Backend — NestJS Services	TypeScript, NestJS, GraphQL, REST	~90,000
Backend — Python Services	FastAPI, LangChain, LangGraph, CrewAI	~70,000
Backend — Go Services	Go, Kafka consumers, dispatch engine	~55,000
Backend — Rust Services	Rust, blockchain interface, telemetry	~35,000
Backend — Java Services	Java, Spring Boot, ERP integrations	~50,000
Backend — Elixir Services	Elixir, Phoenix, WebSocket, Channels	~30,000
Blockchain & Smart Contracts	Solidity, Hardhat, ethers.js, Rust	~25,000
Infrastructure as Code	Terraform, Terragrunt, Helm, ArgoCD	~40,000
CI/CD Pipelines & DevOps	GitHub Actions, Dockerfiles, scripts	~20,000
Test Suites (unit/int/e2e)	Jest, Pytest, Go test, Playwright, Cypress	~110,000
Configuration & Documentation	YAML, JSON, OpenAPI specs, env config	~30,000
TOTAL ESTIMATED		~810,000 LoC

9. Technical Documentation Suite — 19 Markdown Files

Master agentic development ruleset and standards documents.

#	DOCUMENT	CONTENTS
01	MASTER_OVERVIEW.md	System vision, architecture philosophy, module map, platform purpose
02	TECH_STACK.md	Full polyglot stack specification — all languages, frameworks, services
03	FRONTEND_SPEC.md	Next.js vs Angular module allocation, design system, component library
04	BACKEND_MICROSERVICES.md	Service map, language allocation, API contracts, inter-service comms
05	AI_AGENTIC_LAYER.md	LangChain, LangGraph, CrewAI agent designs, HITL protocols
06	BLOCKCHAIN_LAYER.md	Polygon CDK architecture, smart contracts, DID, audit trail spec
07	DATABASE_ARCHITECTURE.md	Supabase, MongoDB Atlas, Upstash Redis, TimescaleDB, Elasticsearch
08	INFRASTRUCTURE_IaC.md	Terraform, AWS EKS, Coolify, Helm, ArgoCD, Karpenter spec
09	OBSERVABILITY.md	Grafana, Prometheus, OTel, Loki, Tempo, Sentry, Datadog, PagerDuty
10	SECURITY_COMPLIANCE.md	POPIA, GDPR, biometrics policy, auth, WAF, penetration testing
11	PAYMENTS_FINANCIAL.md	All 7 payment gateways, smart contract settlements, reconciliation
12	API_INTEGRATION_SPEC.md	All third-party APIs, ERP/POS adapters, partner API standards
13	DEVOPS_CICD.md	GitHub Actions workflows, ArgoCD GitOps, Checkov IaC scanning
14	RESTAURANT_STOREFRONT_SPEC.md	Module 15 full specification — Uber Eats-style partner storefronts
15	DRIVER_ECOSYSTEM_SPEC.md	Driver onboarding, biometrics, wallet, scoring, insurance flow
16	ESG_SUSTAINABILITY.md	EV fleet targets, carbon tracking, solar integration, DFI reporting
17	DEVELOPMENT_ROADMAP.md	300+ phase incremental build plan with milestones and deliverables
18	AGENTIC_DEV_STANDARDS.md	HITL rules, agent standards, best practices for agentic development
19	LOC_ESTIMATION.md	Full LoC breakdown by service, language, module, and test suite

10. Development Roadmap — 300+ Phases

Incremental build plan organised by phase group. Each phase adds atomic, testable functionality.

Phase Groups Overview

PHASE GROUP	PHASES	FOCUS AREA	DELIVERABLE
Foundation	001–020	IaC, repo setup, CI/CD, base Kubernetes cluster	Live AWS EKS cluster + Terraform state
Core Services	021–060	Auth, API Gateway, database schemas, base services	Auth + DB + API skeleton live
Corporate Module	061–090	Next.js corporate site, CMS, investor dashboard	Corporate landing + CMS live
Driver Ecosystem	091–130	Driver onboarding, biometrics, rental, wallet	Driver portal + payment flows live
Fleet Modules	131–160	Contracted fleet, branded fleet, command centre	Fleet management live
Restaurant Storefronts	161–190	Partner storefronts, menu management, ordering	Restaurant platform live
AI & Agents	191–220	LangChain/LangGraph/CrewAI agents, SageMaker model	AI dispatch + ESG agents live
Blockchain Layer	221–240	Polygon CDK, smart contracts, DID, audit trail	Blockchain settlement live
Enterprise Logistics	241–260	Takealot/DSV integrations, SLA enforcement	Enterprise module live
ESG & Compliance	261–275	Carbon tracking, ESG reporting, POPIA audit	ESG dashboard + reports live
Observability	276–285	Full OTel instrumentation, Grafana dashboards	Full observability live
Communications	286–292	Twilio, Firebase, SendGrid integration	Notifications live
Developer Portal	293–298	Public API docs, sandbox, partner key management	Developer portal live
Hardening & Launch	299–310+	Pen testing, load testing, chaos engineering, launch	Production launch

Sample Phases — Foundation Group (001–020)

PHASE	TITLE	MODULE	STACK	OUTPUT
P001	Repository Structure Setup	Foundation	GitHub, monorepo, Nx	Monorepo + workspace config
P002	Terraform AWS Provider Init	Foundation	Terraform, AWS	AWS provider + state backend
P003	VPC + Subnet Architecture	Foundation	Terraform, AWS VPC	Multi-AZ VPC live
P004	EKS Cluster Provisioning	Foundation	Terraform, AWS EKS	Kubernetes cluster live
P005	Helm + ArgoCD Bootstrap	Foundation	Helm, ArgoCD	GitOps CD pipeline live
P006	Coolify on EC2 Setup	Foundation	Coolify, EC2, Docker	Internal deploy UI live
P007	Base GitHub Actions CI	Foundation	GitHub Actions	CI pipeline running
P008	Karpenter Autoscaler Config	Foundation	Karpenter, EKS	Node autoscaling live
P009	Istio Service Mesh Install	Foundation	Istio, Helm, EKS	mTLS + traffic mgmt live
P010	Prometheus + Grafana Stack	Foundation	Prometheus, Grafana	Base metrics dashboard

P011	Loki + Tempo Setup	Foundation	Loki, Tempo, Grafana	Log + trace pipelines live
P012	OpenTelemetry Collector	Foundation	OTel, Kubernetes	OTel instrumentation base
P013	AWS WAF + Shield Config	Foundation	AWS WAF, Shield	DDoS protection live
P014	HashiCorp Vault Deploy	Foundation	Vault, EKS	Secrets management live
P015	Supabase Project Init	Foundation	Supabase, PostgreSQL	DB + auth + realtime live
P016	MongoDB Atlas Cluster	Foundation	MongoDB Atlas	Document store live
P017	Upstash Redis Setup	Foundation	Upstash, Redis	Cache + queues live
P018	AWS MSK Kafka Cluster	Foundation	AWS MSK, Kafka	Event streaming backbone
P019	Snyk + SonarQube CI Hooks	Foundation	Snyk, SonarQube	Security scanning in CI
P020	PagerDuty + Sentry Config	Foundation	PagerDuty, Sentry	Alerting + error tracking

11. Security & Compliance Framework

POPIA-first, GDPR-ready, biometric-compliant security architecture.

CONTROL	IMPLEMENTATION
Authentication	Auth0 (OAuth2, SSO, MFA) + AWS Cognito (API/mobile auth layer)
Secrets Management	HashiCorp Vault (self-hosted on EKS) + AWS Secrets Manager
Encryption at Rest	AES-256 for all database storage; Supabase + MongoDB Atlas native encryption
Encryption in Transit	TLS 1.3 enforced across all services; Istio mTLS between microservices
Biometric Data (POPIA)	Separate biometric data store; explicit consent flows; right-to-erasure API
Facial Recognition	AWS Rekognition — driver identity verification, liveness detection
Fraud Detection	AI anomaly detection via SageMaker; LangGraph fraud investigation agents
API Security	AWS WAF + Shield Advanced; rate limiting via Upstash Redis; API key rotation
Code Security	Snyk (dependency scanning); SonarQube (SAST); OWASP ZAP (DAST)
IaC Security	Checkov scans all Terraform plans in CI before apply
Audit Trail	AWS CloudTrail + blockchain immutable ledger for delivery/payment events
GDPR Compliance	Data residency controls; right to access/erasure; consent management
POPIA Compliance	SA-local data controls; POPI Act Article 8 biometric compliance

12. ESG & Sustainability Framework

Environmental, Social, and Governance framework engineered for DFI compliance.

The ESG framework is a core revenue and compliance asset — not a marketing checkbox. It is architected to satisfy the reporting requirements of sefa, KZN Growth Fund Agency, TIKZN, and impact investors, with automated ESG report generation via CrewAI ESG Agents and real-time carbon tracking via TimescaleDB.

ESG PILLAR	TARGET	MEASUREMENT	REPORTING
Environmental — EV Fleet	30% electric scooters	Per-vehicle energy tracking via IoT	Monthly ESG dashboard
Environmental — Carbon	Real-time CO2 per delivery	TimescaleDB carbon model	Automated CrewAI report
Environmental — Solar	Solar charging stations	Energy generation tracking	Annual sustainability report
Social — Driver Welfare	Insurance per delivery	Naked/Guardrisk API data	Quarterly DFI report
Social — Job Creation	Gig economy formalisation	Driver registration metrics	sefa impact report
Governance — POPIA	100% compliance	Audit log + consent tracking	Annual compliance cert
Governance — Blockchain Audit	Immutable delivery records	Polygon CDK ledger	On-demand DFI audit

13. Platform Vision & Next Steps

Lastmile Gig is not a delivery app. It is the operating system for South Africa's last-mile economy — a full-stack, blockchain-anchored, AI-first, ESG-compliant platform that sits between drivers, restaurants, corporate clients, enterprise logistics operators, and the financial institutions that fund them all.

Built as a subsidiary of the Affiliated Architecture Group (AAG), the platform inherits institutional credibility, established commercial relationships, and a proven track record in the KwaZulu-Natal built environment. The technology architecture is designed for scale — from the KZN last-mile market to pan-African expansion — with every module engineered to production-grade standards from day one.

Immediate Next Steps

1. Generate all 19 Markdown technical documents (agentic development ruleset)
2. Complete the 300+ phase development roadmap in full
3. Begin Phase 001 — Terraform repository and AWS infrastructure setup
4. Regulatory: POPIA compliance framework, CIPC registration, FSCA micro-insurance licensing
5. Partner outreach: restaurant onboarding pipeline, corporate fleet LOIs
6. Investment: DFI outreach package — sefa, KZN Growth Fund, FNB Business Banking

LASTMILE GIG (PTY) LTD · A SUBSIDIARY OF AFFILIATED ARCHITECTURE GROUP

lastmilegig.aagais.co.za · KwaZulu-Natal, South Africa · 2026

CONFIDENTIAL — NOT FOR DISTRIBUTION WITHOUT WRITTEN AUTHORISATION